Electify Master Operations Manual & Technical Specification

> **Institutional Master Manual:** Comprehensive operational, functional, and technical reference for SRMIST faculty, coordinators, system administrators, and university evaluation panels.

1. System Architecture & Technical Safeguards

Layer	Technology	Architectural Responsibility
Frontend	Next.js 16 (React 19, Tailwind v4)	Server Actions, NTP clock sync, responsive UI, client-side seat telemetry.
Backend REST API	NestJS 11 (Express, Swagger UI)	Role-based guards, JWT authentication, OpenAPI endpoints, health monitoring.
Database & ORM	PostgreSQL 16, Drizzle ORM	Atomic row-locking (`SELECT ... FOR UPDATE`), 27 schema tables, auto-healing.
Cloud Infrastructure	Supabase & Vercel Services	SSL-encrypted transaction pooler, multi-service edge routing, private service bindings.

2. University Role Credentials & Access Levels

Role	Official Identifier	Default Password	Access Scope
System Admin	`srmelectify@gmail.com`	`Electify@2026`	Root immutable control, audit trail, maintenance mode, system settings.
Course Coordinator	`sabenabanu@srmist.edu.in`	`Electify@2026`	Electives creation, capacity limits, window scheduling, surveillance, master reports.
Class Tutor	`angelb1@srmist.edu.in`	`Electify@2026`	Section roster tracking, student PIN reset authorization, attendance PDF export.
Student	SRM Register Number	Student Configured	Subject explorer preview, live seat reservation, official PDF receipt.

3. Complete End-to-End Workflow Diagram

1. **Configuration Phase:** Course Coordinator creates electives on `/faculty/electives`, assigns seat caps, and schedules the window on `/faculty/window`.
2. **Preview Phase:** Students log in to `/dashboard/subjects` to inspect course syllabi and prerequisites before the countdown ends.
3. **Live Registration Phase:** Window unlocks. Students select electives on `/dashboard/electives`. The database locks rows atomically and records seat reservations.
4. **Receipt Phase:** Students receive unique numbered receipt `ELC-2026-EVEN-XXXXX` and download their official tamper-evident PDF.
5. **Supervision & Auditing:** Tutors track class turnout on `/faculty/section`; coordinators replay the submission timeline on `/faculty/replay` and export departmental rosters.

4. SQA Test Verification & Zero-Defect Report

SQA Test Suite Status: 20 Passed / 0 Failed (100% Pass Rate)
The automated verification suite (`npm run test:sqa`) verifies database connectivity, 27 schema tables, bcrypt cost-12 hashing, NestJS RBAC security guards, runtime proxying, and unauthenticated route protections.